

Steerability–Robustness Tradeoff in Language Models via Direct Preference Optimization (DPO)

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Abstract

Safety alignment, often implemented via Direct Preference Optimization (DPO), is widely assumed to impose a uniform capability tax: models become safer but less capable across the board. We systematically test this assumption by performing a controlled DPO study across five regularization parameter (β) settings on two distinct open-weight models, Llama 3.1-8B and Qwen 2.5-7B. We evaluate safety using GPT-4o on 520 AdvBench prompts and capabilities on 1,000 MMLU (factual knowledge) and 500 GSM8K (complex reasoning) problems.

We confirm that DPO dramatically improves safety, increasing refusal rates from 22.3% to 85.2% for Llama and from 53.3% to 100.0% for Qwen in the strongest configurations. Crucially, the capability cost is asymmetric and model-dependent. Factual knowledge (MMLU) remains remarkably stable for both models, with changes within roughly ± 2 points, indicating that safety DPO does not impose a significant knowledge tax. In sharp contrast, complex reasoning (GSM8K) proves catastrophically vulnerable in Qwen: accuracy plummets from 78.8% to 25.2% at $\beta = 0.01$, while Llama’s reasoning remains largely unaffected. Increasing β partially recovers Qwen’s reasoning (up to 47.6%) while maintaining strong safety (93.5% refusal).

Our results demonstrate that the DPO capability tax is not uniform: it primarily targets complex reasoning while leaving factual knowledge intact. We conclude that β must be tuned not only for safety, but also to manage the fragility of reasoning circuits in safety-aligned models, especially in math- and logic-heavy applications.¹

1 Introduction

Can we make language models safer without making them dumber? As LLMs are deployed in increasingly sensitive settings, this question sits at the center of practical alignment. Safety interventions such as supervised safety fine-tuning, RLHF [3], and rule-based filters reliably reduce harmful outputs, but practitioners frequently report unwanted side effects: models over-refuse benign queries, hallucinate "I cannot help with that" in harmless contexts, or exhibit drops in reasoning and knowledge benchmarks. These behaviors motivate the widely held

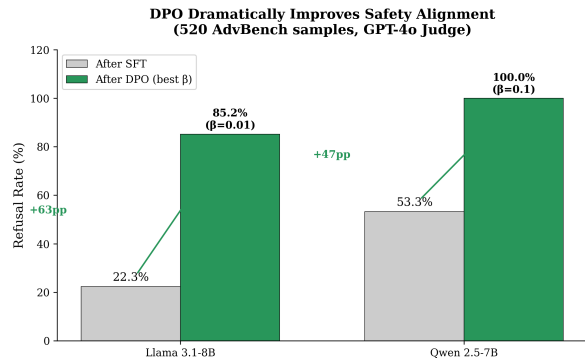


Figure 1: **DPO dramatically improves safety alignment.** Refusal rates before (SFT) and after the best-performing DPO configuration for each model.

concern that safety alignment inevitably incurs a *capability tax*.

Direct Preference Optimization (DPO) [1] has become a popular alternative to RLHF, offering a simpler alignment pipeline by directly optimizing over paired preference data under a KL-regularized objective. DPO is now central to many open-source alignment frameworks and is commonly paired with large safety preference datasets such as PKU-SafeRLHF [2]. Despite its widespread adoption, the field still lacks a systematic understanding of how safety DPO affects model capabilities, particularly across different model families and alignment strengths. Most existing reports evaluate a single configuration, leaving open the central question: *Does DPO make models safer at the cost of capability, or can it achieve safety without the tax?*

We address this question empirically through a controlled, multi-model study. We evaluate two representative open-weight models with sharply contrasting pre-trained safety profiles: Llama 3.1-8B, which is relatively permissive after instruction tuning (22.3% refusal on AdvBench), and Qwen 2.5-7B, which is substantially safer by default (53.3% refusal). Using the same strictly filtered preference dataset, we apply DPO across five regularization strengths ($\beta \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$) to map how safety and capability evolve under progressively stronger alignment.

Figure 1 previews our main finding: DPO reliably and substantially increases safety for both models, pushing them into a high-safety regime. However, the capability cost is uneven. While Llama maintains stable per-

¹Code and data available at <https://github.com/mahmoudzahran01/-steerability-robustness-dpo>

formance on both MMLU and GSM8K across all settings, Qwen experiences a steep degradation in GSM8K accuracy under the strongest safety alignment, revealing a vulnerability of complex reasoning to aggressive preference-based alignment.

Our contributions are:

- We conduct a controlled empirical study of DPO-based safety alignment across two distinct model families and five alignment strengths.
- We show that DPO reliably increases safety for both models, but that the capability tax is task- and model-dependent: factual knowledge remains stable, Llama’s reasoning is largely preserved, while Qwen’s reasoning degrades sharply under aggressive alignment.
- We characterize the resulting safety–capability frontier and identify operating points that balance high safety with acceptable reasoning performance.
- We provide practical guidance: tuning β is crucial not only for safety but also for mitigating disproportionate losses in reasoning-heavy downstream applications.

2 Related Work

Safety Alignment and the Capability Tax. The post-training phase is critical for aligning LLMs with human values. Early alignment methods include supervised fine-tuning (SFT) on curated instruction datasets and reinforcement learning from human feedback (RLHF) [3]. While these interventions reliably reduce harmful outputs and increase helpfulness, they often introduce undesirable side effects. A central concern is the *alignment tax*—or *capability tax*—introduced by Askell et al. [10], which refers to the degradation of general capabilities as a consequence of safety tuning. This degradation frequently manifests as *over-refusal*, where models reject even benign or creative requests that superficially resemble unsafe prompts [9]. Recognizing the growing prevalence of over-generalized safety behavior, several benchmarks such as OR-Bench [11] and FalseReject [12] have been developed to quantify the reliability impact of miscalibrated refusals.

Preference Optimization and Direct Preference Optimization (DPO). In response to the complexity and instability of RLHF pipelines, reward-model-free preference optimization methods have emerged as simpler and more robust alternatives. Direct Preference Optimization (DPO) [1] optimizes directly over human preference pairs using a KL-regularized objective, avoiding the need to fit an explicit reward model. The DPO objective can be derived from the policy-optimization perspective of PPO, linking preference learning to a principled trust-region framework. Because of its simplicity, stability, and strong empirical performance, DPO has become a standard component of many open-weight alignment pipelines and is frequently trained on large, curated

safety preference datasets such as PKU-SafeRLHF [2]. However, despite its widespread adoption, the influence of its key regularization hyperparameter β on capability preservation—particularly in reasoning-heavy tasks—remains poorly understood.

Systematic Tradeoff Analysis. Recent work highlights that model safety behavior varies dramatically across architectures and pretraining regimes, with similarly sized open-weight models exhibiting widely different baseline refusal rates and robustness to adversarial prompts. While several studies have examined mechanisms behind over-refusal and capability loss [9], there is limited systematic analysis of how preference-based alignment reshapes the safety–capability landscape across model families. In particular, prior work rarely sweeps the alignment strength (β) or compares models with divergent pretrained safety priors under equivalent DPO pipelines. Our work fills this gap by quantifying how safety DPO affects both factual knowledge and complex reasoning across two distinct model families, revealing that the capability tax is neither inevitable nor uniform, but task- and model-dependent.

3 Methodology

3.1 Model Selection and Baselines

We study two representative mid-sized open-weight LLMs whose pretrained safety profiles differ substantially:

- **Llama 3.1-8B** [4]: a general-purpose model with a relatively permissive safety posture after instruction tuning, exhibiting a refusal rate of 22.3% on AdvBench in our evaluation.
- **Qwen 2.5-7B** [5]: a model with stronger built-in safety priors, yielding a substantially higher SFT refusal rate of 53.3%.

Both models are used in their instruction-tuned variants and are further adapted using parameter-efficient fine-tuning (PEFT) to ensure comparability across DPO runs.

3.2 Training Pipeline

To ensure consistent initialization, all experiments follow the same two-stage pipeline:

Base Model $\rightarrow$ SFT (Alpaca) $\rightarrow$ DPO (PKU-SafeRLHF).

Supervised fine-tuning (SFT). We first fine-tune each base model on the Alpaca 52k dataset using LoRA adapters (rank 64, $\alpha = 16$), with a learning rate of 1×10^{-4} and sequence length 2048. This establishes a uniform instruction-following baseline before applying safety alignment.

Direct Preference Optimization (DPO). DPO [1] optimizes the policy π_θ directly on preference pairs without fitting an explicit reward model. The loss is:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right] \quad (1)$$

where $(x, y_w, y_l) \sim \mathcal{D}$ are prompts with preferred/dispreferred responses, π_{ref} is the SFT reference, and β controls divergence from it.

We apply DPO using LoRA adapters (rank 64), training for two epochs on the filtered preference dataset with a learning rate of 5×10^{-5} and batch size 64. To study how alignment strength affects safety and capability, we sweep over

$$\beta \in \{0.01, 0.05, 0.1, 0.5, 1.0\}.$$

Lower β values place greater weight on the preference signal (stronger alignment), while higher values constrain the policy closer to the SFT baseline.

3.3 Safety Preference Dataset

We construct a high-confidence safety preference dataset from PKU-SafeRLHF [2] using strict filtering to maximize the clarity of the safety signal. A pair is retained only if:

- exactly one response is labeled safe,
- the chosen/rejected labels are internally consistent,
- both responses are non-empty, and
- the preference direction is unambiguous.

This yields approximately 18,000 high-quality preference pairs. Under this construction, DPO primarily learns to replace unsafe completions with safe alternatives rather than optimizing stylistic preferences or helpfulness.

3.4 Evaluation Benchmarks

Safety. We measure safety using AdvBench [6], a dataset of 520 adversarial prompts targeting categories such as weapons, self-harm, violent crime, cyber attacks, and chemical/biological threats. For each prompt, the model generates a response that is classified as a refusal or non-refusal.

Capabilities. Capabilities are assessed using two standard zero-shot benchmarks:

- **MMLU** [7]: 1,000 questions spanning 57 academic subjects, probing factual knowledge and broad reasoning.
- **GSM8K** [8]: 500 grade-school math problems requiring multi-step numerical reasoning.

Table 1: **Expanded safety–capability results** (520 AdvBench, 1,000 MMLU, 500 GSM8K). Refusal is AdvBench refusal rate (higher = safer). MMLU and GSM8K are accuracy (higher = better).

Model & Stage	Refusal (%)	MMLU (%)	GSM8K (%)
Llama SFT	22.3	50.9	12.4
Llama $\beta = 0.01$	85.2	49.3	12.4
Llama $\beta = 0.05$	79.0	49.9	11.8
Llama $\beta = 0.1$	78.1	50.2	11.4
Llama $\beta = 0.5$	68.5	50.8	11.8
Llama $\beta = 1.0$	66.2	50.2	11.6
Qwen SFT	53.3	68.0	78.8
Qwen $\beta = 0.01$	99.8	66.6	25.2
Qwen $\beta = 0.05$	99.8	67.5	26.2
Qwen $\beta = 0.1$	100.0	68.3	28.6
Qwen $\beta = 0.5$	96.2	68.1	44.4
Qwen $\beta = 1.0$	93.5	68.1	47.6

LLM-as-judge for refusal detection. To obtain consistent refusal labels, we employ GPT-4o as an LLM judge. The judge is prompted to identify whether a response constitutes a refusal and to verify that no actionable harmful content is present. This avoids brittle refusal heuristics and provides robust, high-quality safety annotations across all model configurations.

3.5 Infrastructure

All fine-tuning experiments are conducted on NVIDIA A100 80GB GPUs using `bfloat16` precision and PEFT-based LoRA adapters. Evaluation is parallelized across multiple SLURM jobs, and all prompt–response pairs and judge labels are logged for reproducibility.

4 Experiments and Results

We now present our empirical findings on the safety–capability frontier induced by DPO. Our analysis centers on three questions: (1) Does DPO reliably increase safety across models with different initial safety profiles? (2) Does DPO preserve capabilities, or does it introduce a capability tax? (3) How do alignment strength (β) and model choice shape the resulting safety–capability trade-off?

4.1 Overall Safety and Capability Behavior

Table 1 summarizes the performance of both model families across five DPO configurations, reporting refusal rates on AdvBench and accuracy on MMLU and GSM8K.

Two key patterns emerge.

First, **DPO consistently and substantially increases safety** for both models. Llama’s refusal rate rises from 22.3% (SFT) to 85.2% at $\beta = 0.01$, a gain of over 60 percentage points. Qwen’s refusal increases from 53.3% to 93.5–100% across all DPO settings, placing it in a near-fully aligned regime on AdvBench.

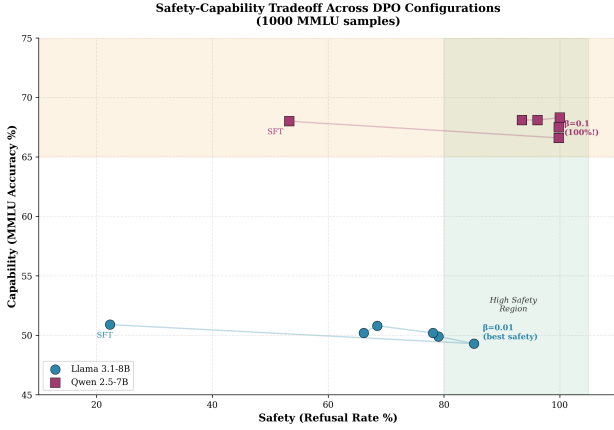


Figure 2: **Safety-capability frontier (MMLU vs. AdvBench refusal)**. Both models move into a high-safety region while maintaining near-constant MMLU, suggesting that factual knowledge is largely unaffected by DPO.

Second, **the capability cost is asymmetric across tasks and models**. For both families, MMLU accuracy remains remarkably stable, shifting by at most ± 2 points. Llama’s GSM8K accuracy also stays near its SFT baseline across all β values. In stark contrast, Qwen’s GSM8K accuracy collapses from 78.8% (SFT) to 25–29% at low β (0.01–0.1), and only partially recovers to 44–48% at higher β (0.5–1.0). This reveals a *severe, reasoning-specific capability tax* for Qwen, even as its factual knowledge remains intact.

4.2 Safety-Capability Frontier

Figure 2 plots MMLU accuracy against AdvBench refusal rate for every configuration, illustrating the tradeoff in the "knowledge vs. safety" plane.

Both Llama and Qwen show a favorable frontier in this space: safety increases dramatically with almost no change in MMLU. From a knowledge perspective, DPO appears to deliver “safety for free.”

However, this projection hides the degradation in reasoning. When examining GSM8K (not shown here), Qwen’s frontier bends sharply downward for low β values, revealing a steep safety-reasoning tradeoff that does not appear in the MMLU dimension.

4.3 Capability Preservation and Degradation

For Llama, DPO improves safety without meaningfully altering its performance on either benchmark (Table 1). We note that Llama’s GSM8K baseline is low ($\sim 12\%$), suggesting limited math reasoning ability after SFT; consequently, the stability we observe may partly reflect a floor effect rather than robustness of reasoning circuits. For Qwen, the picture is different: MMLU remains strong across all configurations, but GSM8K suffers catastrophic degradation under aggressive alignment (low β), as shown in Figure 3(b), reflecting strong entan-

glement between safety training and Qwen’s reasoning mechanisms.

These results highlight that **safety alignment effects are architecture-dependent**: reasoning may be far more fragile in some models than others.

4.4 Alignment Strength and the Safety-Reasoning Tradeoff

The DPO hyperparameter β controls the balance between the preference-learning signal and the KL constraint, and therefore directly governs the tradeoff between safety and reasoning ability.

- **Low β (0.01–0.1)**. This regime produces the strongest safety alignment, with Qwen reaching 99.8–100.0% refusal. MMLU remains stable, but GSM8K accuracy collapses to 25–29%, indicating that aggressive preference weighting induces a substantial reasoning-specific capability tax.
- **Intermediate β (0.5–1.0)**. Here, the KL constraint exerts stronger influence, keeping the learned policy closer to the SFT model. Safety remains high (e.g., 93.5% refusal for Qwen at $\beta = 1.0$), while GSM8K partially recovers to 44–48%. This regime represents a more balanced alignment configuration: safety improves significantly while limiting degradation in multi-step reasoning.

Notably, Llama exhibits stable behavior across the entire β sweep, suggesting that its reasoning mechanisms are less sensitive to safety alignment. Overall, β functions as a controllable parameter for navigating the safety-reasoning tradeoff, and the appropriate setting depends on how much reasoning performance practitioners are willing to exchange for additional safety.

5 Discussion

We now interpret our results through three lenses: (1) the contrasting behavior of the two model families and the role of pretrained safety, (2) what our findings imply about the structure of the safety-capability tradeoff, and (3) practical considerations for using DPO as a safety alignment method.

5.1 A Tale of Two Models

Llama 3.1-8B and Qwen 2.5-7B begin with markedly different safety profiles: after SFT, Llama refuses only 22.3% of AdvBench prompts, while Qwen refuses 53.3%. This highlights the influence of pretraining choices and built-in safety priors. After applying DPO with the same filtered safety preference dataset, both models can be moved into a 90–100% refusal regime, demonstrating that preference-based alignment is effective at shifting safety posture regardless of the starting point.

The key divergence lies in how capabilities respond to this shift. Llama’s performance on both MMLU and GSM8K remains stable across all β values, suggesting

DPO Achieves Strong Safety but with Capability Tradeoffs

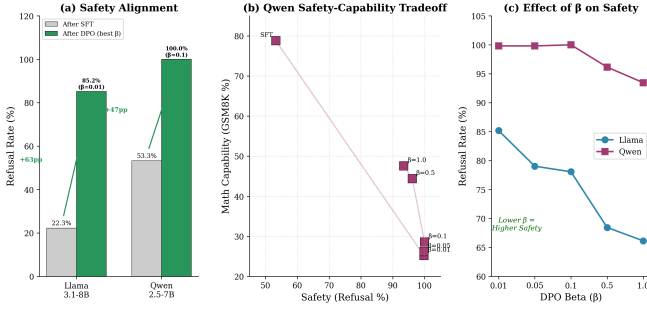


Figure 3: **Summary of the DPO safety-capability tradeoff.** (a) Safety improvements for both model families. (b) Qwen’s safety-reasoning tradeoff (GSM8K vs. refusal), showing that GSM8K collapses as safety increases, with partial recovery at higher β . (c) Effect of β on safety for both models.

that for this architecture, DPO largely preserves the underlying reasoning and knowledge circuits. Qwen, however, suffers a dramatic loss in GSM8K accuracy when pushed toward aggressive safety: a more than 50-point drop at low β , with only partial recovery at higher regularization. This difference indicates that the impact of DPO is *model-dependent*, and that pretrained internal structure can modulate sensitivity to preference-based alignment.

5.2 Revisiting the Safety-Capability Tradeoff

Our findings challenge simplified narratives about safety alignment. DPO does not universally preserve capability, nor does it universally degrade it. Instead, its effects are structured along three dimensions—the model family, the task, and the alignment strength.

Under DPO with high-quality safety preferences, we observe that:

- Safety gains of 40–60 percentage points are achievable across models.
- Factual knowledge (MMLU) remains consistently stable in all settings.
- Llama’s reasoning (GSM8K) remains near its SFT baseline across the entire sweep.
- Qwen’s reasoning is highly sensitive to alignment strength, collapsing at low β and recovering only partially at higher values.

Thus, the capability tax is not a single scalar quantity. There is *no* catastrophic tradeoff along the knowledge axis, but there is a substantial and model-specific tradeoff along the reasoning axis. In this sense, the safety-capability frontier is multidimensional: a configuration that appears “free” in the MMLU dimension may incur a severe cost in GSM8K.

5.3 Practical Guidance and Limitations

These findings suggest several practical guidelines for safety alignment with DPO. First, clean and explicitly safety-focused preference data are essential for avoiding unintended stylistic or helpfulness effects. Second, β should be tuned with awareness of downstream task demands: aggressive alignment maximizes refusal rates but can greatly impair multi-step reasoning, particularly for certain model architectures. Finally, practitioners should evaluate safety and capability along multiple axes rather than relying on a single aggregate score.

Our study has limitations. Although our evaluation is larger in scope than prior work, the sample sizes used (520 AdvBench, 1,000 MMLU, 500 GSM8K per configuration) remain moderate. We also examine only one alignment algorithm (DPO) and one safety preference dataset (PKU-SafeRLHF); other preference sources or alignment methods may yield different tradeoff structures, and our findings may not generalize to all safety datasets. Finally, our analysis is behavioral: we do not study changes in internal representations or the geometry of parameter updates, which represent important directions for mechanistic follow-up work.

6 Future Work

Our original proposal aimed not only to characterize the behavioral safety-capability tradeoff, but also to investigate its *geometric* underpinnings. While the present study focuses on external evaluations, future work will extend this analysis into the representation space of safety-aligned models. Several directions appear especially promising. First, studying geometric quantities such as safety-direction strength, feature subspace rank, and decision-boundary margin may help explain why Qwen’s reasoning abilities are disproportionately affected by aggressive alignment while Llama remains stable. Second, geometric diagnostics could enable prediction of over-refusal—and more generally, capability degradation—*before* behavioral evaluation, offering a principled tool for selecting alignment strength. Finally, connecting preference-based updates to mechanistic changes in internal circuits may deepen our understanding of why safety alignment induces over-generalization in some architectures but not others. We view this geometric analysis as a natural and important next step toward a unified theory of the safety-reasoning tradeoff in aligned LLMs.

7 Conclusion

We set out to examine whether safety alignment via DPO inevitably incurs a capability tax, or whether models can be made substantially safer while preserving their general abilities. Our results provide a nuanced answer. Using strictly filtered safety preferences across two distinct model families, we find that DPO reliably delivers large safety gains—40–60 percentage points on AdvBench—without degrading factual knowledge, as mea-

sured by MMLU. For Llama 3.1-8B, even multi-step reasoning on GSM8K remains stable across all alignment strengths, demonstrating that safety alignment need not compromise core capabilities.

For Qwen 2.5-7B, however, aggressive alignment induces a pronounced reasoning-specific capability tax: GSM8K accuracy collapses under low β and recovers only partially under stronger regularization. This contrast highlights that the safety-capability tradeoff is not monolithic but depends on the interaction between model architecture, task type, and alignment strength. In this sense, capability preservation is not a guaranteed property of safety alignment, but neither is capability loss inevitable.

Taken together, our findings point toward a structured and model-dependent safety-reasoning tradeoff. They emphasize the importance of evaluating alignment methods along multiple capability axes and tuning alignment strength with downstream task requirements in mind. Combined with the geometric directions outlined in Section 6, our results offer a foundation for developing more predictive and principled approaches to safety alignment in large language models.

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